Architecture and Tech Stack Guidelines

Detection and Characterization of Subsurface Ice in Lunar South Polar Regions Using Chandrayaan-2 Radar and Imagery Data for Landing Site and Rover Traverse Planning

1. Guiding Architecture

Use a thin interactive web app over a reproducible geospatial processing pipeline. The demo should not depend on live heavy downloads. Preprocess selected Chandrayaan-2 products into cloud-optimized rasters and compact metadata tables, then serve them through APIs optimized for fast visualization and scoring.

2. System Components

Layer	Recommended Tech	Reason
Frontend	Next.js or Vite React, TypeScript, MapLibre GL, deck.gl	Fast hackathon UI, strong map layer control, good visual polish
API	FastAPI, Pydantic, Uvicorn	Python-native geospatial/ML integration and quick schema validation
Processing	Python, rasterio, rioxarray, xarray, numpy, scikit-image, scikit-learn	Best ecosystem for PDS/geospatial raster transforms
Path planning	networkx or custom grid A*	Simple, explainable route generation over terrain-cost grids
Storage	PostgreSQL + PostGIS, object storage/local files for COG/Zarr	Spatial queries in DB, heavy rasters outside DB
ML/Scoring	scikit-learn baseline, PyTorch optional	Start explainable; add deep learning only if data and time allow
Reports	ReportLab or WeasyPrint	Generate site-selection PDFs from scored outputs

3. Data Pipeline

- Ingest: download PDS4 products from PRADAN/CH2 Map Browse and store raw archives unchanged.
- Parse: read PDS labels, extract raster arrays, geometry, instrument metadata, acquisition time, and projection info.
- Calibrate/derive: generate radar-ratio evidence, roughness proxy, slope, accessibility, illumination/shadow proxy, cold-trap proxy, and crater context layers where possible.
- Normalize: reproject to lunar south-pole stereographic grid and resample to common resolution buckets.
- Tile: export visualization layers as Cloud Optimized GeoTIFFs or MBTiles for fast dashboard rendering.
- Validate: crop NASA PGDA/LRO LOLA products to the TMC-2 AOI and compare regional slope behavior.
- Score: calculate pixel and candidate-site scores, dynamic dashboard metrics, and candidate rankings.
- Route: run A* over accessibility and cold-trap cost layers to connect landing and science points.

4. Processing Architecture

Stage	Inputs	Outputs
Raw archive	PDS IMG/XML/LBL files, product manifests	Immutable raw product registry

Stage	Inputs	Outputs
Geometric processing	SPICE kernels, product footprints, camera/SAR metadata	Projected raster stack
Feature extraction	DFSAR, OHRC, TMC-2 DEM/imagery	Radar evidence, slope, accessibility, morphology, illumination and cold-trap proxies
External validation	NASA PGDA/LRO LOLA slope, roughness, class products	Regional sanity-check metrics and validation focus layer
Decision scoring	Feature rasters and site constraints	Ice evidence confidence, landing safety, traverse cost
Route planning	Accessibility and cold-trap proxy rasters	A* path, relative cost, route evidence notes
Presentation	Scores, layers, paths, explanations	Dynamic dashboard, Judge Demo Mode, mission brief PDF

5. Current Demo Architecture

Capability	Current implementation	Judge value
LOLA validation	Independent NASA PGDA/LRO LOLA slope, roughness, and terrain-class layers are cropped to the TMC-2 AOI.	Shows that the terrain pipeline is being checked against a trusted external reference, not only self-generated layers.
Judge Demo Mode	Seven guided steps walk through the problem, DFSAR evidence, TMC-2 safety, LOLA validation, cold-trap proxy, A* traverse, and final recommendation.	Turns the dashboard into a crisp story that can be presented reliably under time pressure.
A* route	Route from LZ-A to SCI-B is computed from accessibility and cold-trap cost layers instead of being hand drawn.	Connects science target selection to operational rover planning.
Dynamic metrics	Confidence ring, candidate ranking, traverse bars, plan label, and processing-chain text change with the selected layer.	Makes the evidence stack feel alive and auditable instead of static.
Source ledger	Every source is marked as used, validation, context, roadmap, or deprecated with a concrete contribution.	Builds credibility and prevents overclaiming.

6. A* Traverse and Validation Notes

The prototype route from LZ-A to SCI-B is generated with A* over downsampled TMC-2 accessibility and cold-trap proxy rasters. The route has 50 path pixels and a relative cost of 151.74. It is a screening route for judge demonstration and needs geodetic calibration, obstacle inflation, and rover dynamics before mission use.

External validation now uses NASA PGDA/LRO LOLA south-pole slope, roughness, and terrain-class products cropped to the Chandrayaan-2 TMC-2 AOI. The TMC-derived mean slope is 10.67 deg, the LOLA mean slope is 11.63 deg, and the mean difference is 0.96 deg. Because LOLA is 1000 m/pixel, this is a regional sanity check, not meter-scale hazard validation.

7. Deployment for Hackathon

- Local-first demo: Docker Compose with frontend, API, PostGIS, and mounted data directory.
- Preprocessed sample AOIs shipped in repository or external Drive link to avoid live download failures.
- Optional cloud demo: Render/Fly.io for API and Vercel for frontend; use hosted tile files if size permits.
- Keep a fully offline demo mode because mission-data portals and networks may be unreliable during presentation.

8. Development Standards

- Every raster derivative must record source product ID, processing version, projection, resolution, and checksum.

- APIs must return confidence and explanation metadata with every recommendation.

- The UI must separate evidence, inference, and mission recommendation to avoid overclaiming.

- Use configuration files for mission weights so judges can see how priorities change the landing recommendation.

9. Source-to-Evidence Ledger

Source	Status	Contribution
ISRO PRADAN	Used	Official Chandrayaan-2 DFSAR, TMC-2, and OHRC product downloads.
Chandrayaan-2 Map Browse	Used	AOI/product discovery and south-pole footprint selection.
DFSAR science release	Science anchor	CPR/DOP interpretation target; current prototype uses ratio-based candidate evidence.
TMC-2 south-pole DTM/orthobrowse	Used	Slope, accessibility, hillshade, cold-trap proxy, and route cost layers.
OHRC	Context	High-resolution visual hazard context until exact footprint registration is complete.
NASA PGDA/LRO LOLA	Used for validation	Independent 1000 m/pixel slope, roughness, and class sanity check.
USGS ISIS	Roadmap	Production-grade planetary projection/calibration path documented for later hardening.

Source Links

ISRO Science Data Archive - Chandrayaan-2 PRADAN: https://pradan.issdc.gov.in/ch2/

ISSDC Chandrayaan-2 mission data page: https://www.issdc.gov.in/chandrayaan2.html

ISRO DFSAR subsurface-ice release, May 27, 2026: https://www.isro.gov.in/Chandrayaan2_Dual_Frequency_Synthetic_Aperture_Radar.html

Chandrayaan-2 Map Browse: https://chmapbrowse.issdc.gov.in

VEDAS note on downloading Chandrayaan-2 OHRC products: https://vedas.sac.gov.in/static/pdf/SIH_2024/SIH1732_CH2_PS.pdf

NASA Planetary Data System Geosciences Node: https://pds-geosciences.wustl.edu/

NASA PGDA LRO LOLA south-pole products: https://pgda.gsfc.nasa.gov/products/90

USGS ISIS planetary image processing: https://isis.astrogeology.usgs.gov/